**2-1** 试求下列函数的拉氏变换：

(3) $f(t)=\begin{cases}\sin t, & 0\leqslant t\leqslant\pi\\ 0, & t<0,t>\pi\end{cases}$；

(7) $f(t)=\mathrm{e}^{-6t}(\cos 8t+0.25\sin 8t)\cdot 1(t)$；

(8) $f(t)=\mathrm{e}^{-20t}(2+5t)\cdot 1(t)+(7t+2)\delta(t)+\left[3\sin\left(3t-\frac{\pi}{2}\right)\right]\cdot 1\left(t-\frac{\pi}{6}\right)$。

**解：**

(3) $F(s)=\mathrm{L}[\sin t\cdot 1(t)-\sin t\cdot 1(t-\pi)]$

$=\mathrm{L}[\sin t\cdot 1(t)-\sin(t-\pi+\pi)\cdot 1(t-\pi)]$

$=\mathrm{L}[\sin t\cdot 1(t)+\sin(t-\pi)\cdot 1(t-\pi)]=\dfrac{1+\mathrm{e}^{-\pi s}}{s^2+1}$

(7) $F(s)=\mathrm{L}[\mathrm{e}^{-6t}\cos 8t\cdot 1(t)+0.25\mathrm{e}^{-6t}\sin 8t\cdot 1(t)]$

$=\dfrac{s+6}{(s+6)^2+8^2}+\dfrac{2}{(s+6)^2+8^2}=\dfrac{s+8}{s^2+12s+100}$

(8) $F(s)=\mathrm{L}\left\{\mathrm{e}^{-20t}(2+5t)\cdot 1(t)+(7t+2)\delta(t)+\left[3\sin 3\left(t-\frac{\pi}{6}\right)\right]\cdot 1\left(t-\frac{\pi}{6}\right)\right\}$

$=\dfrac{2}{s+20}+\dfrac{5}{(s+20)^2}+2+\dfrac{3\times 3\mathrm{e}^{-\frac{\pi}{6}s}}{s^2+3^2}$

$=2+\dfrac{2}{s+20}+\dfrac{5}{(s+20)^2}+\dfrac{9\mathrm{e}^{-\frac{\pi}{6}s}}{s^2+9}$

**2-2** 试求下列函数的拉氏反变换：

(1) $F(s)=\dfrac{s+1}{(s+2)(s+3)}$；

(4) $F(s)=\dfrac{\mathrm{e}^{-s}}{s-1}$；

(6) $F(s)=\dfrac{4}{s^2+s+4}$；

**解：**

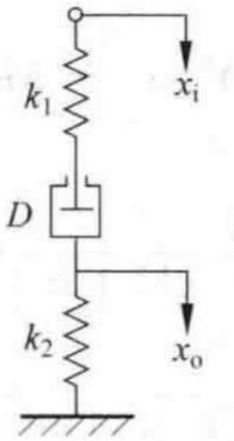
(1) $f(t)=\mathrm{L}^{-1}\left(\dfrac{-1}{s+2}+\dfrac{2}{s+3}\right)=(-\mathrm{e}^{-2t}+2\mathrm{e}^{-3t})\cdot 1(t)$

(4) 用延时定理和衰减定理

$$f(t)=\mathrm{L}^{-1}\left(\frac{\mathrm{e}^{-s}}{s-1}\right)=\mathrm{e}^{t-1}\cdot 1(t-1)$$

(6) $f(t)=\mathrm{L}^{-1}\left[\dfrac{\frac{8\sqrt{15}}{15}\times\frac{\sqrt{15}}{2}}{\left(s+\frac{1}{2}\right)^2+\left(\frac{\sqrt{15}}{2}\right)^2}\right]=\dfrac{8\sqrt{15}}{15}\mathrm{e}^{-\frac{t}{2}}\sin\dfrac{\sqrt{15}}{2}t\cdot 1(t)$

**2-9** 试求图 2.11(a)～(g)所示各机械系统的传递函数。




(b)

**解：**

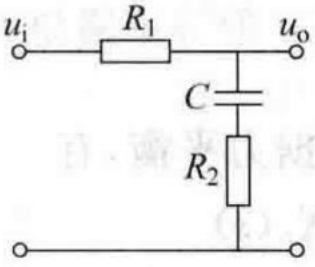
（b）阻尼器的等效弹性刚度为 $Ds$，根据力平衡，有

$$\frac{k_1 Ds}{k_1 + Ds}[X_i(s) - X_o(s)] = k_2 X_o(s)$$

整理得

$$\frac{X_o(s)}{X_i(s)} = \frac{k_1 Ds}{(k_1 + k_2)Ds + k_1 k_2}$$

**2-10** 试求图 2.13(a)～(c)所示各无源电路网络的传递函数。

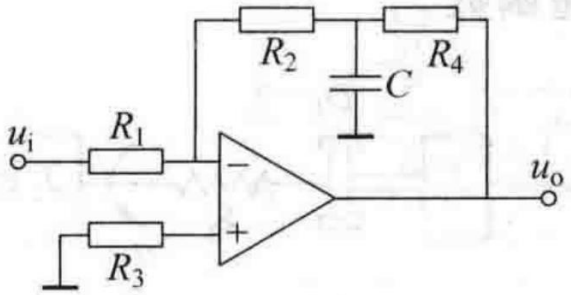



(a)

**解：**

$$\text{(a)}\ \frac{U_o(s)}{U_i(s)} = \frac{\frac{1}{Cs} + R_2}{R_1 + \frac{1}{Cs} + R_2} = \frac{R_2 Cs + 1}{(R_1 + R_2)Cs + 1}$$

**2-11** 试求图 2.14(a)～(d)所示各有源电路网络的传递函数。




(c)

**解：**

（c）设电容 $C$ 和电阻 $R_2$ 连接点的电位为 $u_A$，则有

$$\begin{cases} \dfrac{U_A(s)}{\dfrac{1}{Cs}} + \dfrac{U_A(s)}{R_2} = \dfrac{U_o(s) - U_A(s)}{R_4} \\ \dfrac{U_A(s)}{R_2} = -\dfrac{U_i(s)}{R_1} \end{cases}$$

消去 $U_A(s)$，得

$$\frac{U_o(s)}{U_i(s)} = -\frac{R_2 + R_4}{R_1}\left(\frac{R_2 R_4}{R_2 + R_4}Cs + 1\right)$$